

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Improved risk-adjusted returns by 35% with a cost-aware position smoother on the desk's futures signal.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Wrote a feature store for 120 microstructure features with point-in-time joins, which the research team reused in two later signal projects.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Normalized every feature with the full-sample mean and variance, then split the data into walk-forward folds to validate the signal.
- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2021 - Aug 2021

- Designed the department's entire statistical computing infrastructure, now used by every lab in the faculty.
- Derived a variance bound for a sparse regression estimator; my proof tightens the previous bound by a log factor and is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation with shrinkage and factor models, downloaded 3,000 times in its first year.

Sunrise Bakery | Assistant Store Manager | Metro City, USA | Jun 2020 - May 2021

- Managed opening shifts and a team of 6 bakers and cashiers, keeping the store within its weekly labour budget.
- Ran daily stock counts and supplier orders, cutting unsold bread from 12% to 7% of production.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Tested significance with Diebold-Mariano tests across the 30 indices; the forecast gain held at the 5% level on 24 of them and in both crisis periods.
- Wrote up the method, robustness checks and results as a 12-page working paper, which was presented at the department's financial econometrics seminar.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Engineered features and trained gradient-boosting models for a market prediction competition.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Added 0.4 to the Sharpe ratio of a futures book with an order-book imbalance signal, tested over 18 months out of sample.

SKILLS

Programming: Python, R, SQL, C++, Kafka

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch